not give in to wear and tear easily. Not only are the test equipment being ruggedised, but also test cables are being improved for better operation. Test cables for improved measurements are a necessity but are often forgotten. Of late, cables are being ruggedised and being made safe for operation at high frequencies up to 40GHz.

VNA test cables from Pasternack can withstand production testing for 50-ohm communication systems. Torsion-resistant connector heads are directly attached to the steel conduit cycle to ruggedise the design for up to 75,000 flexure cycles.

A word of caution for optoelectronics. With the optoelectronics time-domain measurement method, scattering parameter measurements on planar waveguides up to 500GHz with 500MHz frequency spacing are easily achievable. In addition to high-frequency measurements, this could also be used to characterise high-frequency coaxial devices and to help realise a precise voltage pulse standard. You need to ensure that you remove the effects of the optical to electrical converter of the photo diode by de-embedding these. This is done after the traditional calibration at the network analyser reference plane to remove the effects of RF cables and connectors.

USB testing made easy. Windows based operating systems have enabled development of applications for automated gain-compression measurements, mixer measurements, IMD measurements and measurements on balanced device architectures. Also, with the advent of USB based power sensors, network analysers can be used with USB sensors as power meters as well.

More recently, femtosecond lasers are making a major difference. Network analysers today offer a higher dynamic range with tuned measurements and excellent receiver linearity. This makes femtosecond lasers a viable option for use in network analysers.

Femtosecond lasers are here

For accurate characterisation of a high-frequency device, a directional coupler is used to separate forward and backward propagating signals. After research showed that a frequency-resolved scattering parameter could also be realised using laser based measurement techniques, the femtosecond laser began to be utilised. But, how?

The femtosecond laser uses a relatively less complex and less expensive femtosecond laser source to generate short, precise voltage test pulses, which travel along a short gold strip built on a gallium arsenide chip.

The pulse's electric field changes the gallium-arsenide's refraction index, so that another laser beam can track and measure phase and amplitude as the signal travels down the strip. It can resolve signals travelling both up and down the conductor, to measure the signal reflected by the material in the circuit.